

# Janarthanan S

Vellore, Tamil Nadu

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## Summary

Fresh B.E. graduate (May 2026) and software engineer with strong foundations in Data Structures & Algorithms, Object-Oriented Design, and scalable backend development. Proficient in Python with hands-on experience building production-ready REST APIs using FastAPI, PostgreSQL, Docker, and CI/CD pipelines. Solved **500+ DSA problems** on LeetCode demonstrating deep algorithmic problem-solving and time/space complexity analysis skills. Passionate about learning new technologies and building innovative software solutions that scale.

## Technical Skills

- **Languages:** Python, Java (Basic), SQL, HTML, CSS
- **Backend & APIs:** FastAPI, REST API Development, CRUD Operations, Pagination, Sorting, Filtering, Pydantic Validation
- **Databases:** PostgreSQL, SQLite, SQLAlchemy ORM, Alembic, Redis (Basic), Database Indexing, Query Optimization
- **Authentication:** JWT, OAuth2, bcrypt
- **Cloud & DevOps:** AWS (EC2, S3, RDS, IAM, CloudWatch), Docker, Docker Compose, Git, GitHub, GitHub Actions (CI/CD)
- **Testing & Tools:** Pytest, Postman, Swagger UI, pgAdmin, VS Code
- **ML & Data:** Pandas, NumPy, OpenCV, YOLOv11, NLP, Deep Learning, Matplotlib
- **CS Concepts:** OOP, DSA, OS Fundamentals, Computer Networks (TCP/IP, HTTP/HTTPS, DNS), DBMS (ACID, Normalization, Indexing), Basic System Design, Distributed Systems (Basic)

## Internship Experience

### Proxenix Software Company

July 2025 – September 2025

*Software Engineer Intern — Backend Systems & ML (Remote)*

- Developed a sentiment analysis system using Python and NLP techniques (TF-IDF, text preprocessing) to classify customer and movie reviews as positive or negative with high accuracy.
- Designed and implemented FastAPI REST APIs for real-time sentiment prediction, integrating the trained machine learning model with client applications.
- Built an end-to-end ML pipeline covering data ingestion, text preprocessing, feature extraction, model training, and evaluation for efficient sentiment prediction.
- Collaborated in sprint reviews and documented technical findings, improving alignment between engineering work and stakeholder expectations.

## Projects

### Smart Hostel Issue Tracking System | Python, FastAPI, SQLAlchemy, Docker, Pytest, GitHub Actions | Live Demo Feb 2026

- Reduced inefficiencies in manual hostel grievance handling by building a secure backend with FastAPI, PostgreSQL, SQLAlchemy ORM, Alembic migrations, JWT authentication, and Docker.
- Implemented RESTful CRUD APIs with pagination, filtering, Pydantic validation, Pytest unit tests, and GitHub Actions CI/CD to improve reliability and deployment efficiency.
- Analyzed complaint data to identify recurring infrastructure issues, enabling faster maintenance decisions through data-driven insights.

### Pothole Detection & Real-Time Alert System | Python, YOLOv11, OpenCV, Deep Learning

Nov 2025

- Addressed unreliable road hazard detection by developing a real-time pothole detection system using YOLOv11 and OpenCV.
- Improved model performance through dataset augmentation and hyperparameter tuning, achieving **92% precision** while reducing false positives.

### Aljust Spoken English Platform | Python, NLP, Firebase, REST API, Cloud Backend | Live Demo

Jan 2026

- Addressed the lack of personalized English learning by developing an AI-powered platform using NLP, Firebase, and REST APIs.
- Built a scalable cloud backend with real-time synchronization, improving user engagement by **40%** through personalized learning experiences.

## Education

### Madras Institute of Technology, Anna University

Sep 2023 – May 2026

*B.E. Electronics and Communication Engineering – CGPA: 7.28/10*

*Chennai, Tamil Nadu*

### Rajagopal Polytechnic College

2020 – 2023

*Diploma in Electronics and Communication Engineering – 94.30%*

*Vellore, Tamil Nadu*

## Achievements & Certifications

- Solved **500+ DSA problems** on LeetCode (Python) — strong algorithms, OOP design, and problem-solving skills.
- Grew **LinkedIn community to 2,000+ followers** sharing Python, backend, and software development content.
- **HackerRank Certifications:** Python (Basic) | SQL (Basic) | Problem Solving (Basic) | REST API (Intermediate)
- **Python for Beginners** – Scaler Academy | **Basics of Python** – Udemy

## Leadership & Collaboration

- Led cross-functional engineering teams of 4–6 members; managed task allocation, sprint planning, and on-time delivery ensuring high-quality software releases.
- Mentored peers in Python, DSA, and algorithmic problem-solving, fostering a collaborative and growth-oriented engineering environment.